

Arduino Based WINDOWS PC VOLUME REMOTE CONTROL

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Presented here is a project that turns your Arduino into a remote control. Infrared (IR) remotes have been very prominent since 1980's. In today's world almost everything is controlled by a remote. This project allows you to control the master volume of a Windows PC, a television set, set-top box, home theatre system and the like.

Standard data formats

There are two common data formats for IR control: RC5 coding and NEC coding.

RC5 code. RC5 code uses a bi-phase coding, and the carrier frequency is fixed at 36kHz. Transmission of the data begins with two start bits followed by a toggle bit (Fig. 1). Toggle bit changes its value at each new key press. Five address

bits represent the address of the device to be controlled. Six command bits contain the information to be transmitted.

Each bit in the data consists of half a bit period with no transmission and half a bit period with a burst of 32 pulses at 36kHz. The most suitable IR receivers for receiving RC5 code are those with AGC2 setting and a band-pass frequency of 36kHz. Some examples are TSOP1236, TSOP4836, TSOP34836, TSOP39236 and TSOP36236.

NEC code. NEC code uses bursts at a carrier frequency of 38kHz. It starts transmission using a leader code, a burst with a length of 9ms, followed by a pause of 4.5ms and then the data (Fig. 2). The original purpose of this leader code was to let the internal control loops in the receiver modules to settle.

After transmitting the data, only the leader code and a single bit are transmitted repeatedly for as long as a key is pressed. A special property of this code is a constant word length in combination with pulse distance modulation.

Address and data bits are transmitted twice, first as a normal byte, followed by an inverted byte. The half period burst portion of each bit contains 22 pulses, each with a width of 8.77µs and a period of 26.3µs. 0 is represented by a pulse distance of 1.125ms, and 1 by a pulse distance of 2.25ms.

Eight address bits are used to identify the device to be controlled. Further eight bits are used for transmission of command data. As mentioned above, the words are always followed, without a pause, by inverted words. For example,

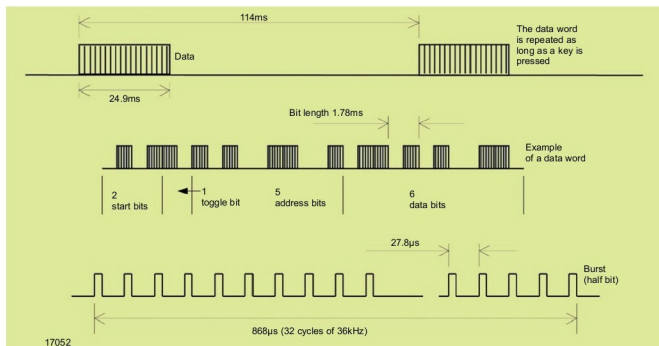


Fig. 1: RC5 transmission code (Image courtesy: www.vishay.com)